Gray means part of a previous submission (50 clubs, 14.5 hours)

Problem 1 (IITJEE)

We have,

$$4^x + 6^x = 9^x$$

$$\Leftrightarrow 1 + \left(\frac{3}{2}\right)^x = \left(\frac{3}{2}\right)^{2x}$$

We let $y = \left(\frac{3}{2}\right)^x$.

$$\Leftrightarrow 1 + y = y^2$$

By the quadratic formula,

$$y = \frac{1 \pm \sqrt{5}}{2}$$

Clearly, $y = \left(\frac{3}{2}\right)^x < 0$ is impossible, so we take the positive solution.

$$y = \frac{1+\sqrt{5}}{2} \Leftrightarrow x = \log_{\frac{3}{2}} \frac{1+\sqrt{5}}{2}$$

Problem 2 (CMIMC 2018 A5)

Notice that,

$$bc + \frac{1}{a} = \frac{1}{a+b+c} \Leftrightarrow abc = \frac{a}{a+b+c} - 1$$

Similarly,

$$abc = \frac{b}{a+b+c} - 2 = \frac{c}{a+b+c} - 7$$

Adding these up, we get,

$$3abc = -10 + \frac{a+b+c}{a+b+c} \Leftrightarrow abc = -3$$

Now, we multiply the equations together to get,

$$\left(bc + \frac{1}{a}\right)\left(ca + \frac{2}{b}\right)\left(ab + \frac{7}{c}\right) = (abc)^2 + 10abc + 23 + \frac{14}{abc} = 9 - 30 + 23 + \frac{14}{-3} = \frac{1}{(a+b+c)^3}$$

Therefore, $a + b + c = -\frac{\sqrt[3]{3}}{2}$.

Problem 3 (EGMO 2019/1)

We claim the only solutions are the permutations of $(a, b, c) = (0, 1, 1)$ and $(a, b, c) = (0, -1, -1)$, and also $(a, b, c) = \left(\frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}\right)$ or $(a, b, c) = \left(-\frac{\sqrt{3}}{3}, -\frac{\sqrt{3}}{3}, -\frac{\sqrt{3}}{3}\right)$. Each of these obviously works.

Using $ab + bc + ca = 1$,

$$a^2b + c = b^2c + a = c^2a + b$$

$$\Leftrightarrow a^2b + abc + bc^2 + ac^2 = b^2c + a^2b + abc + a^2c = ac^2 + ab^2 + b^2c + abc$$

Taking the first equality, we cancel to get,

$$bc^2 + ac^2 = b^2c + a^2c$$

For $c = 0$, we get $ab = 1$; $a^2b = a = b$ so $a = b = \pm 1$.

This gives the permutations of $(a, b, c) = (0, 1, 1)$ and $(a, b, c) = (0, -1, -1)$. If not,

$$\Longrightarrow bc + ac = b^2 + a^2$$

Similarly, we get,

$$ab + ac = b^2 + c^2;$$

$$ab + bc = a^2 + c^2$$

Adding these up, we get,

$$\Leftrightarrow 0 = (a - b)^2 + (b - c)^2 + (c - a)^2$$

Therefore, $a = b = c$. From the given, we get the last solution, where $(a, b, c) = \left(\frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}\right)$ and $(a, b, c) = \left(-\frac{\sqrt{3}}{3}, -\frac{\sqrt{3}}{3}, -\frac{\sqrt{3}}{3}\right)$.

Problem 4 (Vietnam 2014/1)

We claim that $x_k = \sin\left(\frac{60}{2^k}\right)$; $y_k = \cos\left(\frac{60}{2^k}\right)$. We prove this by induction,

Base case: $k = 1$, already given.

Inductive step: $k = n$ to $k = n + 1$,

$$x_{n+1}^2 + y_n = 2$$

$$\Leftrightarrow x_{n+1}^2 + 2\cos\left(\frac{60}{2^n}\right) = 2$$

$$\Leftrightarrow x_{n+1}^2 = 2 - \left(2 - 4\sin^2\frac{60}{2^{n+1}}\right) = 4\sin^2\frac{60}{2^{n+1}}$$

Since x_{n+1} is positive,

$$\Leftrightarrow x_{n+1} = 2\sin\left(\frac{60}{2^{n+1}}\right)$$

Now,

$$x_{n+1}y_{n+1} - x_n = 0$$
$$\Leftrightarrow 2\sin\left(\frac{60}{2^{n+1}}\right)y_{n+1} = 2\sin\left(\frac{60}{2^n}\right)$$
$$\Leftrightarrow 2\sin\left(\frac{60}{2^{n+1}}\right)y_{n+1} = 4\sin\left(\frac{60}{2^{n+1}}\right)\cos\left(\frac{60}{2^{n+1}}\right)$$

Therefore, $y_{n+1} = 2\cos\left(\frac{60}{2^{n+1}}\right)$ as well.

Conclusion: Induction complete for all $k \geq 1$.

Therefore,

$$\lim_{n\to\infty} x_n = 0; \lim_{n\to\infty} y_n = 2$$

<u>Problem 5 (IMO 2014/1)</u>

For the sake of contradiction, let $a_n < \frac{a_0+a_1+\cdots+a_n}{n} \leq a_{n+1}$ be not satisfied for all $n \geq 1$.

We claim that $\frac{a_0+a_1+\cdots+a_n}{n} > a_{n+1}$ for all $n \geq 1$. We prove this by induction.

Base case: $n = 1$

Plugging in $n = 1$ to the inequality, we do not have,

$$a_1 < \frac{a_0 + a_1}{1} \leq a_2$$

However, clearly $a_1 < \frac{a_0+a_1}{1}$ as a_0 is positive. Therefore, for the inequality to be not satisfied,

$$\frac{a_0 + a_1}{1} > a_2$$

Inductive step: $n = k - 1$ to $n = k$

Plugging in $n = k$ to the inequality, we do not have,

$$a_k < \frac{a_0 + \cdots + a_k}{k} \leq a_{k+1}$$

However,

$$a_k < \frac{a_0 + \cdots + a_k}{k} \Leftrightarrow \frac{(k-1)a_k}{k} < \frac{a_0 + \cdots + a_{k-1}}{k} \Leftrightarrow a_k < \frac{a_0 + \cdots + a_{k-1}}{k-1}$$

By the inductive hypothesis this is true. Therefore, $\frac{a_0+\cdots+a_k}{k} \leq a_{k+1}$ must be false for the inequality to be not satisfied. We hence have,

$$\frac{a_0 + \cdots + a_k}{k} > a_{k+1}$$

Conclusion: By induction the claim is proven.

For any k, we have,

$$\frac{a_0 + \cdots + a_k}{k} > a_{k+1}$$

We now fix $a_0 = x$.

$$\frac{a_0 + \cdots + a_k}{k} > a_{k+1}$$

$$\Leftrightarrow \frac{a_0}{k} + \frac{a_1 + a_2 + \cdots + a_k}{k} > a_{k+1}$$

$$\Leftrightarrow \frac{a_0}{k} > a_{k+1} - \frac{a_1 + a_2 + \cdots + a_k}{k}$$

However, since a is strictly increasing, $a_{k+1} \geq a_i + (k + 1 - i)$. Therefore,

$$\Rightarrow \frac{a_0}{k} > a_{k+1} - \frac{(a_{k+1} - k) + \big(a_{k+1} - (k-1)\big) + \cdots + (a_{k+1} - 1)}{k}$$

$$\Leftrightarrow \frac{a_0}{k} > a_{k+1} - a_{k+1} + \frac{1 + 2 + \cdots + k}{k}$$

$$\Leftrightarrow \frac{a_0}{k} > \frac{k+1}{2}$$

Taking k large, this is impossible.

+5 clubs; 1 hour

Problem 6 (14AIME14)

Sketch:

Move -4 over and add 1 to each fraction, then divide by x.

Substitute $y = x - 11$, $z = y^2 - 50$ to get apart from $x = 0, y = 0$, we also have two more solutions for z.

Maximal solution ends up being $x = 11 + \sqrt{50 + 10\sqrt{2}}$.

Problem 7 (EGMO 2015/4)

We claim the answer is $n = 5$, which can be constructed with $(a_1, a_2, a_3, a_4, a_5) = (477, 7, 29, 35, 43)$.

We let $x_n = a_{n+1} - a_n$. We know x_n is always positive for $n \geq 2$ since square roots are positive. Since a is increasing, x_n is also increasing for $n \geq 3$.

Notice that,

$$x_{n+1}^2 - x_n^2 = a_{n+1} - a_{n-1} = x_n + x_{n-1}$$

$$\Leftrightarrow x_{n+1} - x_n = \frac{x_n + x_{n-1}}{x_{n+1} + x_n}$$

For $n \geq 4$, the RHS is less than 1. However, $x_{n+1} - x_n \geq 1$, so there's a contradiction. Hence, the maximum length of the sequence is 5.

Problem 8 (24OIME6)

Sketch: Graph it. The graph is a bunch of segments from $(m, 0)$ to $(m+1, m)$ on $[m, m+1)$ for all integers m. Clearly $k \geq 1$ gives infinite solutions on negatives, so $k < 1$ and we don't have any positive solutions x. Having at least 24 solutions means $y = kx$ passes strictly above $(-22, -23)$ so the slope is strictly less than $\frac{23}{22}$. Similarly, at most 24 solutions means passing below or at $(-23, -24)$ so the slope is greater than or equal to $\frac{24}{23}$. The answer is hence,

$$\frac{1}{l} = \frac{1}{\frac{23}{22} - \frac{24}{23}} = 506$$

Problem 9 (06AIMEII15)

Note that $x, y, z > 0$. In addition, $x = \sqrt{y^2 - \frac{1}{16}} + \sqrt{z^2 - \frac{1}{16}} < y + z$ and similar results means that x, y, z can be the side lengths of a triangle.

By the Pythagorean Theorem, we can have x, y, z be the side lengths of a triangle with altitudes of length $\frac{1}{4}$ to x, $\frac{1}{5}$ to y, and $\frac{1}{6}$ to z.

Using $A = bh$, we can write $x = 4k$, $y = 5k$, $z = 6k$. We use Heron's to get,

$$A = \frac{15k^2\sqrt{7}}{4} = \frac{k}{2} \Leftrightarrow k = \frac{2\sqrt{7}}{105}$$

Therefore, $x = \frac{8\sqrt{7}}{105}; y = \frac{2\sqrt{7}}{21}; z = \frac{4\sqrt{7}}{35}$.

Problem 10 (20BWAY5)

We claim the only answer is $x = y = z = 0$ which obviously works.

Using the quadratic formula on the first equation,

$$x = \frac{\pm\sqrt{-y^3z - z^2y}}{y} \Leftrightarrow (xy)^2 = -y^3z - z^2y$$

$$y = \frac{-x^2 \pm \sqrt{x^4 - 4z^3}}{2z} \Leftrightarrow (2yz + x^2)^2 = x^4 - 4z^3$$

$$z = \frac{-y^2 \pm \sqrt{y^4 - 4x^2y}}{2} \Leftrightarrow (2z + y^2)^2 = y^4 - 4x^2y$$

We hence have,

$$z^3 + z^2y + zy^3 + x^2y = \frac{1}{4}(x^4 + y^4)$$

$$\Leftrightarrow 4z^3 + 4z^2y + 4zy^3 + 4x^2y - x^4 - y^4 = 0$$

$$\Leftrightarrow -4(xy)^2 - (2yz + x^2)^2 - (2z + y^2)^2 = 0$$

This is only possible when,

$$xy = 2yz + x^2 = 2z + y^2 = 0$$

We focus on $xy = 0$. If $x = 0$,

$$y^2z + z^2 = 0;$$

$$4z^3 + 4z^2y + 4zy^3 = y^4$$

If $z = 0$ also, then $y = 0$. So, in the case that $z \neq 0$, from the first equation,

$$y^2 + z = 0$$

However, this contradicts with $2z + y^2 = 0$ If $z \neq 0$. Therefore, $x = 0 \Longrightarrow y = 0, z = 0$.

For $y = 0$, we plug in to get,

$$z^2 = 0 \Leftrightarrow z = 0$$

$$0 = \frac{1}{4}x^4 \Leftrightarrow z = 0$$

This also gives the $x = y = z = 0$ answer. Therefore, there are no other answers.

+11 clubs; 2.5 hour

<u>Problem 11 (24AIMEII11)</u>

We claim that all triples that satisfy $a + b + c = 300$ where $a = 100, b = 100$, or $c = 100$ work.

Note that,

$$a^2b + a^2c + b^2a + b^2c + c^2a + c^2b = 6 \cdot 10^6$$

$$\Leftrightarrow \sum_{cyc} a^2(b+c) = 6 \cdot 10^6$$

$$\Leftrightarrow \sum_{cyc} a^2(300-a) = 6 \cdot 10^6$$

We plug in $A = a - 100; B = b - 100; C = c - 100$. This way, $A + B + C = 0$ and,

$$\Leftrightarrow 6 \cdot 10^6 = \sum_{cyc} (A+100)^2(200-A)$$

$$\Leftrightarrow 0 = \sum_{cyc} A(30000 - A^2)$$

$$\Leftrightarrow 0 = A^3 + B^3 + C^3$$

Plugging in $C = -A - B$,

$$\Leftrightarrow 0 = -AB(A+B) = ABC$$

Therefore, either $a = 100, b = 100$, or $c = 100$. To verify that this works, WLOG let $A = 0$,

$$A + B + C = 0 \Leftrightarrow C = -B$$

We plug this into $0 = A^3 + B^3 + C^3$ and it is indeed correct.

The solutions are permutations of $(a,b,c) = (0,100,200), (1,100,199), \ldots, (99,100,101)$ for 600 solutions in total or $(a,b,c) = (100,100,100)$. In total, there are 601.

Problem 12 (MRJ479)

We let,

$$x = \frac{a^2}{bc} - 1; y = \frac{b^2}{ac} - 1; z = \frac{c^2}{ab} - 1$$

Therefore,

$$\Leftrightarrow x^3 + y^3 + z^3 = 3xyz$$
$$\Leftrightarrow (x+y+z)(x^2+y^2+z^2-xy-yz-zx) = 0$$

Case one: $x + y + z = 0$, meaning,

$$\frac{a^2}{bc} + \frac{b^2}{ac} + \frac{c^2}{ab} = 3$$
$$\Leftrightarrow (a+b+c)(a^2+b^2+c^2-ab-bc-ca) = 0$$

Note that,

$$a^2+b^2+c^2-ab-bc-ca = 0$$
$$\Leftrightarrow \frac{(a-b)^2}{2} + \frac{(b-c)^2}{2} + \frac{(c-a)^2}{2} = 0$$

We get, $a = b = c$ which is not allowed.
Therefore, we must have $a + b + c = 0$ in this case.

Case two: $x^2 + y^2 + z^2 - xy - yz - xz = 0$
In this case, we similarly end up with $x = y = z$, so,

$$\frac{a^2}{bc} - 1 = \frac{b^2}{ac} - 1 = \frac{c^2}{ab} - 1$$
$$\Leftrightarrow a^3 = b^3 = c^3$$
$$\Leftrightarrow a = b = c$$

Once again, this is not allowed, so this case is impossible.

The only possibility results in $a + b + c = 0$.

Problem 13 (13AIMEII15)

Adding the first two equations,

$$\cos^2 A + \cos^2 C + 2\cos^2 B + 2\sin B \sin(A + C) = \frac{15}{8} + \frac{14}{9}$$

$$\Leftrightarrow \cos^2 A + \cos^2 C + 2 = \frac{15}{8} + \frac{14}{9}$$

$$\Leftrightarrow \cos^2 A + \cos^2 C = \frac{103}{72}$$

We let $\cos^2 C + \cos^2 A + 2\sin C \sin A \cos B = x$. We may analogously get,

$$\cos^2 A + \cos^2 B = x - \frac{4}{9};$$

$$\cos^2 B + \cos^2 C = x - \frac{1}{8}$$

Hence,

$$\cos^2 C = \frac{7}{8}; \cos^2 A = \frac{5}{9}; \cos^2 B = x - 1$$

Since A, C are acute, their cosines are positive. Therefore, $\cos C = \frac{\sqrt{14}}{4}; \cos A = \frac{\sqrt{5}}{3}$.

We then have $\sin^2 C = \frac{1}{8}; \sin^2 A = \frac{4}{9}$. Since they are positive, $\sin C = \frac{\sqrt{2}}{4}; \sin A = \frac{2}{3}$.

However, we have,

$$x = 1 + \cos^2 B = 1 + \cos^2(A + C)$$

$$= 1 + (\cos A \cos C - \sin A \sin C)^2$$

$$= 1 + \left(\frac{\sqrt{2}}{6} - \frac{\sqrt{70}}{12}\right)^2 = \frac{37}{24} - \frac{\sqrt{35}}{18}$$

+16 clubs; 4.5 hours

Problem 14 (ZCB390FF)

By AM-GM,

$$a_{50}+b_{50}>20 \Longleftarrow \sqrt{a_{50}b_{50}}>10 \Leftrightarrow a_{50}b_{50}>100$$

Let $p_n = a_n b_n$. We have,

$$p_2 = a_2 b_2 = \left(a_1+\frac{1}{b_1}\right)\left(b_1+\frac{1}{a_1}\right) = a_1 b_1 + \frac{1}{a_1 b_1} + \frac{a_1}{b_1} + \frac{b_1}{a_1} \geq 4$$

For $n \geq 2$,

$$p_{n+1} = \left(a_n+\frac{1}{b_n}\right)\left(b_n+\frac{1}{a_n}\right) = p_n + \frac{1}{p_n} + \frac{a_n}{b_n} + \frac{b_n}{a_n} > p_n + \frac{a_n}{b_n} + \frac{b_n}{a_n} \geq p_n + 2$$

Therefore, $p_3 > p_2 + 2; p_4 > p_3 + 2; \ldots$ We induct to get $a_{50}b_{50} > 100$ as desired.

Problem 15 (08GER33)

We let $a = x+y; b = x-y$. We have,

$$a^2 b = 675;$$

$$b\left(\frac{a^2+b^2}{2}\right) = 351$$

$$\Leftrightarrow a^2 b + b^3 = 702$$

$$\Leftrightarrow b^3 = 27$$

$$\Leftrightarrow b = 3; a = \pm 15$$

For $(a,b) = (15,3)$, we have $x = 9; b = 6$. For $(a,b) = (-15,3)$, we have $x = -6; y = -9$.

+21 clubs; 6 hours

Problem 16 (21ARMLOCI10)

Decreasing a_n by x will decrease a_{n-1} by $\frac{x}{n}$ such that the equality is still satisfied. Then, by induction, it will decrease a_1 by $\frac{x}{n\cdot(n-1)\cdot\ldots\cdot 2} = \frac{x}{n!}$.

We let $a'_{2021} = (2021!+1)^2 = a_{2021} - 2020!$ with a' sequence following the same rules as a. We have, $a'_1 = a_1 - \frac{2020!}{2021!}$.

We claim that $a'_n = (n!+1)^2$ for all $1 \leq n \leq 2021$. We use induction in reverse to prove this.

Base case: $n = 2021$. We are given this.

Inductive step: $n = k \rightarrow n = k-1$

Conclusion: By induction $a_1' = (1! + 1)^2$.

We have $a_1' = 4 = a_1 - \frac{2020!}{2021!} \Leftrightarrow a_1 = 4 + \frac{1}{2021} = \frac{8085}{2021}$.

Problem 17 (20SIME14)

Sketch: We can plug in $x = 2y + 1$ since $2y + 1$ is bijective to get the same number of answers. We would then get, $p(x) = 8y^3 - 6y + 1$. Clearly, $y = 0$ is a solution always. We may then plug in $y = \cos(\theta)$ for some complex number θ to exploit triple angle formula. We then put this on the unit circle to have, for some positive $x \neq 0$ (WLOG let $a > b$),

$$x^{2\cdot 3^a} + 1 = x^{3^b + 3^a} + x^{3^a - 3^b}$$

$$\Leftrightarrow \left(x^{3^a - 3^b} - 1\right)\left(x^{3^a + 3^b} - 1\right) = 0$$

There are $3^a - 3^b$ positive roots with a bijection to y. In total, $3^a - 3^b + 1$ solutions for y for some $a > b$. Case-working on b, there are 15 solutions so $999 - 15 = 984$ non-solutions.

Problem 18 (IMO 2018/2)

We claim the answer is all the multiples of 3, which can be achieved with the sequence -1, -1, 2, -1, -1, 2, …

To prove that all other n fail, note that,

$$a_i a_{i+1} + 1 = a_{i+2} \Leftrightarrow a_i a_{i+1} a_{i+2} = a_{i+2}^2 - a_{i+2};$$

$$a_{i+1} a_{i+2} + 1 = a_{i+3} \Leftrightarrow a_i a_{i+1} a_{i+2} = a_i a_{i+3} - a_i$$

Summing $a_{i+2}^2 - a_{i+2} = a_i a_{i+3} - a_i$ cyclically, we get,

$$\sum_{cyc} a_{i+2}^2 - a_{i+2} = \sum_{cyc} a_i a_{i+3} - a_i$$

$$\Leftrightarrow a_1^2 + a_2^2 + \cdots + a_n^2 = a_1 a_4 + a_2 a_5 + \cdots + a_{n-2} a_1 + a_{n-1} a_2 + a_n a_3$$

If n is not a multiple of 3, we take indices mod 3 to get,

$$\Leftrightarrow a_1^2 - a_1 a_4 + a_4^2 - a_4 a_7 + a_7^2 + \cdots = (a_1 - a_4)^2 + (a_4 - a_7)^2 + \cdots = 0$$

We end up getting $a_1 = a_2 = \cdots = a_n$. We solve $x^2 + 1 = x$ to get there's no solution.

+28 clubs; 8 hours

Problem 19 (20EGMO2)

Since y is a permutation of x, $\sum_{i=1}^{2020} y_i^3 = \sum_{i=1}^{2020} x_i^3$. Therefore,

$$\sum_{i=1}^{2020} \left((x_i+1)(y_i+1)\right)^2 = 8\sum_{i=1}^{2020} x_i^3$$

$$\Leftrightarrow \sum_{i=1}^{2020} \left((x_i+1)(y_i+1)\right)^2 = 4\sum_{i=1}^{2020} x_i^3 + 4\sum_{i=1}^{2020} y_i^3$$

We claim that,

$$\left((x_i+1)(y_i+1)\right)^2 \geq 4x_i^3 + 4y_i^3$$

If $y_i = x_i - 1$,

$$\Leftrightarrow \left(x_i^2 + x_i\right)^2 \geq 4x_i^3 + 4(x_i-1)^3$$

$$\Leftrightarrow x_i^4 + 2x_i^3 + x_i^2 \geq 8x_i^3 - 12x_i^2 + 12x_i - 4$$

$$\Leftrightarrow x_i^4 - 6x_i^3 + 13x_i^2 - 12x_i + 4 \geq 0$$

$$\Leftrightarrow (x-1)^2(x-2)^2 \geq 0$$

This is true with equality cases at $x_i = 1, 2$. Analogously, if $y_i = x_i + 1$, the inequality is also true except with equality cases at $x_i = 0, 1$.

For $y_i = x_i$,

$$\Leftarrow (x_i+1)^4 > 8x_i^3$$

This is clearly true for $x_i \geq 8$. We plug in other values of x to get this is true for all non-negative x_i. Therefore, for,

$$\sum_{i=1}^{2020} \left((x_i+1)(y_i+1)\right)^2 = 4\sum_{i=1}^{2020} x_i^3 + 4\sum_{i=1}^{2020} y_i^3$$

If we add together the inequalities $\left((x_i+1)(y_i+1)\right)^2 \leq 4\left(x_i^3 + y_i^3\right)$ over all i, we have that each of the inequalities must take the equality case. Therefore, we must have, for every pair (x_i, y_i), either $(0,1)$, $(1,0)$, $(1,2)$, or $(2,1)$.

We must have, for some $1 \leq j \leq 2020$ and positive n,

$$x_1 = x_2 = \cdots = x_j = n-1;$$

$$x_{j+1} = x_{j+2} = \cdots = x_{2020} = n$$

Therefore, in the first two cases, $n = 1$ and in the second two cases $n = 0$. In the first case, note that there's for ever 1 in $x_1, \ldots, x_{2020}, y_1, \ldots, y_{2020}$, it is paired to a 0. Therefore, there's an equal amount of both. Since y is a permutation of x, there is then an equal amount of 1s and 0s in $x_1, \ldots, x_{2020}$. This is possible if and only if,

$$x_1 = \cdots = x_{1010} = 0; x_{1011} = \cdots = x_{2020} = 1$$

In which case, taking $y_i = x_{2021-i}$ suffices. Similarly, the second two cases give $n = 2$ and,

$$x_1 = \cdots = x_{1010} = 1; x_{1011} = \cdots = x_{2020} = 2$$

Problem 20 (Iberoamerican 2021/4)

WLOG, let $1 = a^2 + x^2 = \cdots$. We let, complex numbers,

$$z_1 = a + xi; z_2 = b + yi; z_3 = c + zi$$

We have,

$$1 = |z_1| = |z_2| = |z_3| = |z_1 + z_2| = |z_2 + z_3| = |z_3 + z_1|$$

If we draw the unit circle ω_1 on the complex plane and another unit circle ω_2 centered at z_1. We let z_3, z_4 be the intersections of ω_1, ω_2, we see that $z_2 = z_3 - z_1$ or $z_2 = z_4 - z_1$, so z_2 is 120 degrees clockwise or counter-clockwise from z_1.

The same applies to all the others, so z_1, z_2, z_3 are all in 120 degree apart. Therefore, so are z_1^2, z_2^2, z_3^2.

$$z_1^2 + z_2^2 + z_3^2 = 0$$

$$\Longrightarrow Re(z_1^2 + z_2^2 + z_3^2) = 0$$

$$\Leftrightarrow a^2 + b^2 + c^2 = x^2 + y^2 + z^2$$

Problem 21 (23IMC2)

Note that,

$$B^3 = (ABC + 2\,\mathrm{id})$$

$$\Longrightarrow B^6 = A^2B^2C^2 + 4\,ABC + 4\,\mathrm{id}$$

Using $A^2 = B^2 = C^2$,

$$\Leftrightarrow B^6 = B^6 + 4\,ABC + 4\,\mathrm{id}$$

$$\Leftrightarrow ABC = -\mathrm{id}$$

$$\Longrightarrow A^2B^2C^2 = \mathrm{id}$$

$$\Leftrightarrow A^6 = \mathrm{id}$$

All problems completed :D

New earnings: +46 clubs; 10 hours

Previous total: 50 clubs; 14.5 hours (some problems later removed)